

DAEGWALLYEONG, Korea, Republic of

WMO: 471000

Lat: 37.6772N

Lon: 128.7183E

Elev: 2539

StdP: 13.40

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.3	3.8	-13.3	2.9	5.9	-10.1	3.5	8.9	33.4	16.0	30.2	17.7	9.9	270	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	81.1	70.0	78.7	68.9	76.4	67.9	72.7	78.0	71.2	75.9	70.1	74.5	9.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	125.9	75.6	69.8	120.6	74.1	68.6	115.4	73.0	38.3	78.2	36.9	76.2	35.8	74.6	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.4	23.8	20.9	-6.0	84.9	4.5	2.0	-9.3	86.3	-11.9	87.5	-14.4	88.6	-17.7	90.1		
			-6.9	74.5	4.3	1.7	-10.0	75.7	-12.6	76.7	-15.0	77.7	-18.1	78.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	19.4	23.7	32.8	44.7	54.7	61.2	68.1	68.2	58.6	48.5	36.5	23.9
	DBStd	18.13	7.52	7.68	7.81	7.53	6.10	5.42	5.44	5.21	5.48	5.96	7.57	7.91
	HDD50	3760	949	738	534	193	26	1	0	0	3	99	409	808
	HDD65	7546	1414	1158	998	610	322	135	34	30	204	512	856	1273
	CDD50	1984	0	0	1	34	172	337	561	564	261	52	2	0
	CDD65	295	0	0	0	0	3	21	131	129	11	0	0	0
	CDH74	1078	0	0	0	5	38	115	436	454	30	0	0	0
	CDH80	113	0	0	0	1	4	9	48	50	1	0	0	0
Wind	WSAvg	8.6	10.9	9.8	9.9	10.0	9.0	6.4	7.7	6.4	5.3	7.1	9.5	10.9
Precipitation	PrecAvg	66.3	2.3	2.0	2.8	3.9	4.3	6.5	11.6	15.0	9.0	4.3	3.0	1.6
	PrecMax	96.9	9.6	6.2	7.4	9.6	8.0	13.8	27.4	30.9	26.5	16.4	10.2	7.4
	PrecMin	45.1	0.2	0.0	0.6	0.8	1.2	1.0	4.3	4.9	0.0	0.7	0.3	0.2
	PrecStd	16.1	2.6	1.7	1.9	2.1	2.1	3.5	6.1	7.6	6.5	4.3	2.6	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	49.8	59.5	72.4	78.8	80.9	84.1	84.2	77.6	68.9	59.2	46.8
		MCWB	34.4	43.3	45.9	54.5	57.7	62.9	72.4	72.3	66.3	56.9	49.1	40.6
	2%	DB	36.3	42.8	54.3	66.5	73.5	77.2	81.1	81.2	73.6	64.5	54.9	42.0
		MCWB	31.4	36.0	42.7	50.1	56.0	62.6	70.8	71.8	64.8	53.9	47.7	37.0
	5%	DB	33.4	38.6	49.6	62.4	70.1	74.4	78.6	78.8	70.6	61.4	51.8	38.5
		MCWB	29.6	32.6	40.3	48.1	55.1	62.1	69.6	71.0	63.7	53.1	45.1	33.7
	10%	DB	30.9	35.3	45.6	58.5	67.1	71.7	76.2	76.4	67.9	58.6	48.6	35.5
		MCWB	27.4	30.2	37.5	46.9	53.9	61.9	69.2	69.8	62.6	52.5	43.2	31.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	45.0	48.5	58.8	62.8	69.0	74.4	75.1	69.7	61.5	53.5	42.1
		MCDB	38.9	46.5	55.5	67.7	69.8	73.7	80.3	80.6	74.5	62.9	55.8	45.2
	2%	WB	32.6	36.4	44.7	53.6	60.1	66.8	72.7	73.1	67.2	57.5	49.7	37.7
		MCDB	34.8	41.0	51.3	62.6	67.3	71.9	77.8	78.7	70.7	61.0	53.7	40.6
	5%	WB	30.3	33.3	41.3	50.4	58.0	65.1	71.1	71.8	65.3	55.5	46.5	34.5
		MCDB	32.8	37.8	48.6	59.1	65.6	70.7	75.8	76.7	68.9	58.7	49.8	37.6
	10%	WB	27.8	30.9	38.1	47.9	56.0	63.5	70.0	70.6	63.4	53.7	43.9	32.0
		MCDB	30.4	34.6	44.8	56.2	64.1	69.1	74.7	75.0	66.8	57.1	47.5	34.9
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.6	16.1	18.3	18.1	14.7	10.1	10.9	13.9	17.2	15.7	15.6
		MCDBR	17.8	19.6	21.4	25.5	24.7	21.2	14.4	14.7	18.5	21.7	20.3	17.7
		MCWBR	13.7	13.8	13.1	12.9	11.2	9.5	6.1	6.9	10.3	12.5	13.2	13.2
	5% WB	MCDBR	14.0	17.2	19.4	20.2	19.0	15.2	12.1	12.8	14.2	15.5	15.5	15.8
		MCWBR	12.0	13.2	13.3	12.8	11.3	8.6	6.3	7.0	9.2	10.8	12.2	13.0
Clear-Sky Solar Irradiance	taub	0.369	0.425	0.512	0.549	0.553	0.599	0.566	0.506	0.439	0.431	0.399	0.362	
	taud	2.056	1.935	1.784	1.760	1.792	1.746	1.898	2.071	2.195	2.085	2.096	2.113	
	Ebn at Noon	246	246	236	237	239	227	234	246	256	242	233	239	
	Edh at Noon	39	50	64	70	69	72	62	50	42	43	37	35	
All-Sky Solar Radiation	RadAvg	778	1020	1303	1551	1731	1629	1333	1334	1230	1088	771	693	
	RadStd	67	97	134	123	163	133	191	175	151	89	100	60	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	+0.61	N/A	N/A	N/A	+0.95	+0.76	N/A	N/A	+180	N/A
(47)	Regional (39 neighbors)	N/A	N/A	-1.57	N/A	N/A	N/A	N/A	N/A	+176	+97

Nomenclature: See separate page